Laboratory 03 Report: PCA and LDA

Machine Learning and Pattern Recognition

1 PCA analysis

What do you observe from the PCA projected histograms? What are the effects on the class distribution?

Applying Principal Component Analysis (PCA) transforms the dataset by projecting it onto a new set of orthogonal axes ordered by their variance. Observing the histograms of the projected features, the first PCA direction (Direction 1) captures the maximum overall spread of the dataset. As we move towards Direction 6, the variance noticeably decreases.

However, the effect on the class distribution reveals an interesting property for this dataset. Along the first PCA direction, the "Fake" (label 0) and "Genuine" (label 1) classes are quite effectively separated. Although PCA is unsupervised and maximizes the total variance of the data regardless of class labels, for this specific data, the direction of maximum data spread aligns with the direction of maximum class discrimination. This causes the two classes to separate notably along the first principal component.

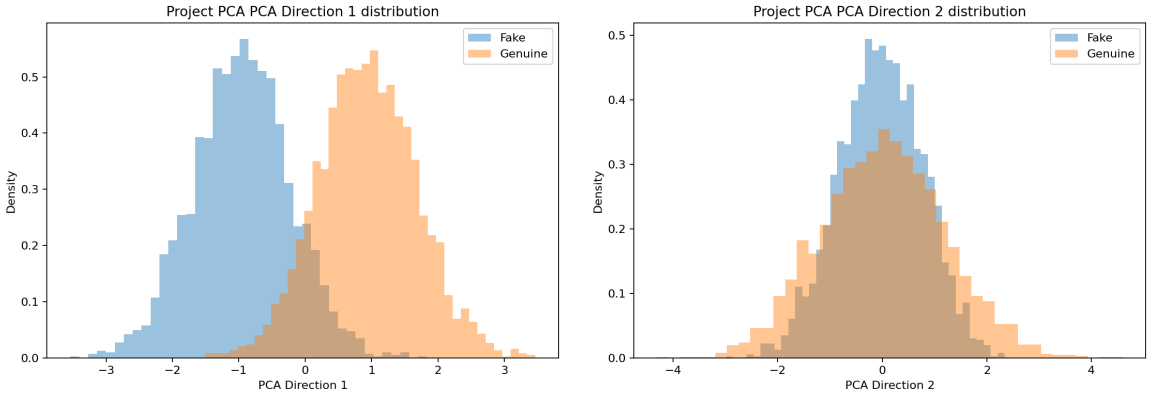


Figure 1: Histograms of the data projected onto the first two PCA directions. Notice the effective separation between the Fake and Genuine classes along the first PCA direction.

Can you spot the different clusters inside each class? Have the clusters really changed? Have the classes really become "more separated"?

Yes, by projecting onto the first principal component, the classes have become separated in that 1D subspace, highlighting the difference between them.

Applying PCA and retaining all $m = 6$ dimensions acts mathematically as a rigid orthogonal rotation of the original feature space. Because an orthogonal transformation perfectly preserves Euclidean distances (L_2 norm), any two data points that were overlapping or close together in the raw feature space remain exactly the same distance apart in the PCA-transformed space. We are simply viewing the exact same geometric structure from a newly rotated coordinate system, where the first axis now clearly shows the separation.

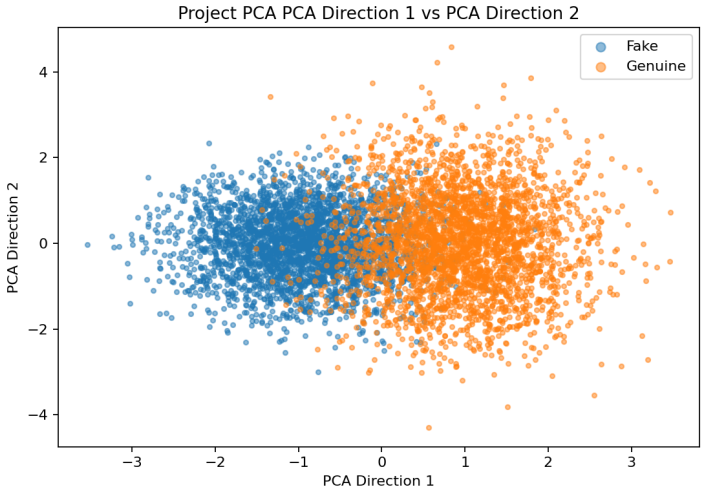


Figure 2: Scatter plot of the first two PCA directions. The geometric clusters of the original dataset are rotated so that the classes become separated along the first dimension.

This confirms that exploratory analyses (like viewing 1D and 2D shadows of a 6D object) can sometimes reveal the global separation between classes even with an unsupervised method. Because the direction of maximum variance aligns with the class separation, PCA behaves similarly to a supervised approach for this dataset. Applying Linear Discriminant Analysis (LDA), which specifically targets the direction that maximizes the distance between class means while minimizing intra-class variance, reveals a separation very similar to that achieved by the first PCA direction.

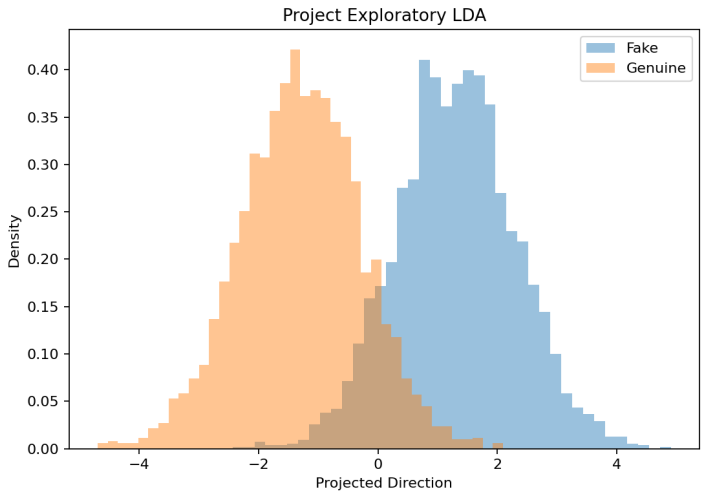


Figure 3: LDA Projection (1D). The supervised LDA projection clearly separates the density centers of the Fake and Genuine classes, achieving a result similar to the first PCA direction.

2 LDA analysis

Apply LDA and compute the histogram of the projected samples. What do you observe? Do the classes overlap over the found direction?

Observing the histogram of the samples projected onto the single LDA direction, we see a stark contrast to unsupervised methods like PCA. The data forms a clearly bimodal distribution with the "Fake" and "Genuine" classes occupying distinct regions of the projected space. However, the classes do still overlap over the found direction. The right tail of the "Fake" class intersects with the left tail of the "Genuine" class in the center. This indicates that while linear separability is significantly improved, a perfectly clean linear boundary that yields 0% error does not exist for this dataset.

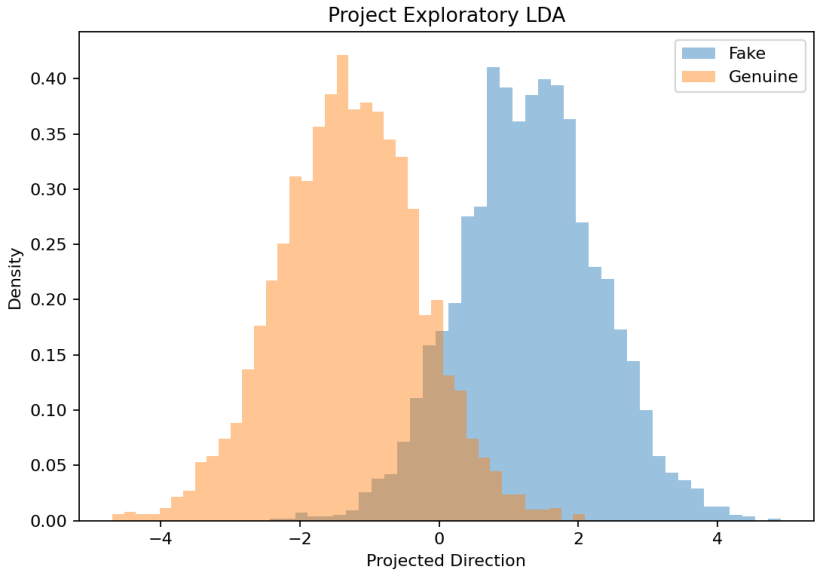


Figure 4: Histogram of the data projected onto the LDA direction. The density centers are separated, but the tails overlap.

Compared to the histograms of the 6 features you computed in Laboratory 2, is LDA finding a good direction with little class overlap?

Yes, LDA is finding an exceptionally good direction compared to the raw features. In Laboratory 2, the histograms of the 6 individual features showed massive overlap; the distributions for "Fake" and "Genuine" were often nearly superimposed (especially for features 1 and 2), meaning no single raw feature could act as a strong linear classifier. By taking a linear combination of all features, LDA successfully finds a 1-dimensional subspace where the class overlap is drastically minimized.

Try applying LDA for classification... Compute the predictions on the validation data, and the corresponding error rate.

We divided the dataset into a model training set and a validation set (containing 2000 samples). We applied LDA and oriented the projection so that the mean of the "Genuine" class was larger than the "Fake" class. Using the default threshold—the simple average of the two projected training class means, which was calculated as approximately -0.0185—the model achieved an error rate of **9.3%** on the validation data.

Now try changing the value of the threshold. What do you observe? Can you find values that improve the classification accuracy?

By sweeping the threshold across the range of projected values and evaluating the validation error at each step, we observe that the error rate forms a distinct curve with a clear minimum. The default geometric threshold (the midpoint of the projected means) does not yield the absolute lowest error.

We can indeed find values that improve classification accuracy. By shifting the threshold slightly lower to approximately -0.1080, the validation error rate improves, dropping from 9.3% down to **9.1%**. This observation suggests that the two class distributions are not perfectly symmetric or do not share the exact same variance; therefore, a slightly biased decision boundary discriminates the overlapping tails more effectively than the exact geometric midpoint.

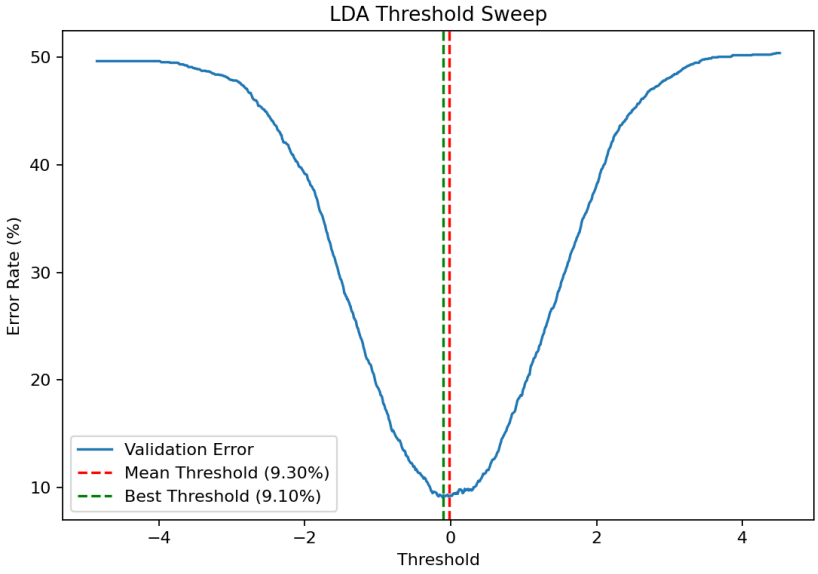


Figure 5: Validation error rate as a function of the classification threshold. The green dashed line indicates the empirical optimum, which achieves a slightly lower error rate (9.1%) than the default geometric mean threshold (9.3%, red dashed line).

3 PCA + LDA analysis

Apply PCA as a pre-processing step for LDA. Evaluate the error rate for different values of PCA dimensions ($m = 1, \ldots, 6$). What do you observe?

To evaluate the impact of PCA as a pre-processing step, we first project the data onto an m-dimensional PCA subspace, and subsequently apply LDA to reduce it to a single dimension for classification. The default threshold (average of the projected means) is used for evaluation on the validation set.

As shown in Table 1, varying the number of PCA dimensions (m) has virtually no impact on the classification performance. When we keep all dimensions ($m = 6$), the error rate perfectly matches the baseline pure LDA model at 9.30%. Aggressively reducing the dimensions down to $m = 2, 3$, or 4 yields a negligible shift to 9.25%, and reducing it to a single dimension ($m = 1$) yields 9.35%. Overall, the results are roughly identical across the board.

PCA Dimensions (m)	Validation Error Rate (%)
1	*9.35*
2	*9.25*
3	*9.25*
4	*9.25*
5	*9.30*
6	9.30

Table 1: Classification error rates using PCA as a pre-processing step for LDA across different subspace dimensions.

Does PCA pre-processing improve the classification accuracy? Why or why not?

No, PCA pre-processing does not significantly alter the classification accuracy for this dataset; the error rates remain virtually identical regardless of the chosen dimensionality m.

To deeply understand why PCA does not impact the LDA classification outcome, we must trace the mathematical relationship between their optimization objectives step-by-step.

Step 1: The Core Optimization Objectives
Linear Discriminant Analysis (LDA) is a supervised method that seeks to find a projection that maximizes class separability. It does this by maximizing the between-class scatter matrix (S_B) while minimizing the within-class scatter matrix (S_W). Conceptually, it optimizes a ratio proportional to:

$$\text{LDA Objective} \propto \frac{S_B}{S_W}$$

Principal Component Analysis (PCA), conversely, is an unsupervised method. It is entirely blind to class labels and seeks only to maximize the total overall variance of the dataset, represented by the total scatter matrix (S_T).

Step 2: The Scatter Matrix Identity
The fundamental mathematical link between these two algorithms is the Total Scatter Identity:

$$S_T = S_B + S_W$$

This identity dictates that the total variance in the dataset (S_T) is always exactly equal to the sum of the variance between the classes (S_B) and the variance within the classes (S_W).

Step 3: The Uniformity Constraint
For this specific dataset, empirical observation implies that the within-class variance (S_W) is

relatively stable and isotropic (uniform in all directions). Mathematically, this means S_W acts roughly as a constant scaling factor across the feature space ($S_W \approx cI$, where c is a constant).

Step 4: Objective Convergence
If we substitute this uniformity constraint into the Total Scatter Identity, the equation becomes:

$$S_T \approx S_B + \text{Constant}$$

When PCA searches for the directions of maximum total variance ($\arg\max S_T$), it is evaluating $\arg\max(S_B + \text{Constant})$. Because the constant term does not affect the location of the mathematical maximum, PCA inadvertently finds the exact directions that maximize the between-class variance (S_B). Under this specific geometric condition, the unsupervised PCA objective mathematically aligns with the supervised LDA objective.

Step 5: The Redundancy of Dimensionality Reduction
Because both algorithms are ultimately aligning with the same fundamental variations (the separation between the class means), the dimensions with the lowest total variance—which are discarded by PCA—contain only background noise. They contain absolutely zero discriminative information. When LDA is subsequently applied to this PCA-reduced subspace, it is provided the exact same optimal discriminant directions it would have found by searching the raw, uncompressed feature space. Consequently, dropping dimensions via PCA removes no valuable information, making the pre-processing step mathematically redundant for classification purposes.